Constraint inequalities

1 A warehouse is stocked with two types of storage containers: square boxes and rectangular boxes. The square boxes cost $\$55$ each and have a volume of $2\ \text{m}^3$ and the rectangular boxes cost $\$65$ each and have a volume of $7\ \text{m}^3$. The warehouse has a total storage space of $200\ \text{m}^3$ and there is a budget of $\$800$ to purchase the containers.

Let x represent the number of square boxes purchased, and y represent the number of rectangular boxes purchased.

a If it's not possible for a number of boxes to be less than zero, construct constraint inequalities for x and y to represent this fact.

b Write an inequality relating x and y for the total budget for the warehouse.

c Write an inequality relating x and y for the total storage space of the warehouse.

2 A bicycle manufacturer employs a mechanic and a painter to construct two types of bikes: racing bikes and mountain bikes. The time each worker spends on each bike is shown below (assume both workers can work on the same bike at the same time).
Let x represent the number of racing bikes built and y represent the number of mountain bikes built.

Each worker can work at most 40 hours in a week.

	Mechanic	Painter
Racing Bike	5 hours	10 hours
Mountain Bike	8 hours	4 hours

a Write an inequality relating x and y to the total time spent working by the mechanic.

b Write an inequality relating x and y to the total time spent working by the painter.

3 A light bulb manufacturer produces two types of light bulbs: incandescent and fluorescent.

Let x represent the number of incandescent bulbs produced in a day, and y represent the number of fluorescent bulbs produced in a day.

a If it's not possible for a number of lightbulbs to be less than zero., construct constraint inequalities for x and y to represent this fact.

b Up to 200 light bulbs can be produced in one day. Write an inequality to represent this.

c The final inequality representing constraints on the number of light bulbs that can be produced each day is:

$$y \geq \frac{x}{2}$$

Interpret this final constraint in context.

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4 A company produces cans of cat food and dog food. Each product passes through two machines in their production, and the amount of time each machine takes to process one pallet of pet food is shown in the table below.
Each machine can operate for 12 hours in a day.

	Machine A	Machine B
Cat Food	2 hours	3 hours
Dog Food	4 hours	1.5 hours

Let x represent the number of pallets of cat food processed each day, and y represent the number of pallets of dog food processed each day.

a Write a constraint inequality for Machine A.

b Write a constraint inequality for Machine B.

c Can the values of x and y be negative? Explain your answer.

d Hence, write two more constraint inequalities.

e Graph the region defined by the constraint inequalities for Machine A and Machine B on a cartesian plane.

5 Lamborgotti Motors produces two types of car: a sedan and a coupe. Each type of car requires assembly in two different factories. The time that each car is required to be in each factory is given in the table below.
Each factory can operate for 18 hours a day.

	Factory A	Factory B
Sedan	3 hours	4.5 hours
Coupe	2 hours	1.5 hours

Let x represent the number of sedans produced each day, and y represent the number of coupes produced each day.

a Write a constraint inequality for Factory A.

b Write a constraint inequality for Factory B.

c Construct the constraint inequalities for x and y given the number of guitars cannot be a negative number.

d Graph the region defined by the constraint inequalities for Factory A and Factory B

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6 Mahindra produces two types of tractors: an All Purpose and a Harvester. Each type of tractor requires production in two different factories. The time that each tractor is required to be in each factory is given in the table below.
Each factory can operate for 18 hours a day.

Let x represent the number of All Purpose tractors produced each day, and y represent the number of Harvesters produced each day.

	Factory A	Factory B
All Purpose	4 hours	6 hours
Harvesters	2 hours	1.5 hours

a Wite a constraint inequality for Factory A.

b Write a constraint inequality for Factory B.

c Construct the constraint inequalities for x and y given the number of guitars cannot be a negative number.

d Graph the region defined by the constraint inequalities for Factory A and Factory B.

7 Eileen is spending the week at a sports camp. She is able to attend coaching sessions for her two favourite sports: basketball and football. She can attend a maximum of 26 sessions and a minimum of 20 sessions per week.

Eileen's basketball coach has told her that she must attend at least 5 basketball sessions. Her football coach has told her that she must go to at least 8 more football sessions than basketball sessions.

Let x represent the number of basketball sessions she attends, and y represent the number of football sessions she attends.

a From this information, construct the set of four constraint inequalities.

b Graph the region defined by the four constraint inequalities.

c Hence, list the four vertices of the feasible region.

d Using the four vertices determine the maximum number of basketball sessions that Eileen can attend.

8 Edward is going to a music camp for one week. Edward plays drums and guitar and can attend lessons for both instruments. He can attend a maximum of 18 lessons and a minimum of 13 lessons per week.

Edward's guitar teacher has told him that he must not go to more than two times as many guitar lessons as drum lessons. His drum teacher has told him he must attend at least five drumming lessons.

Let x represent the number of drum lessons he attends, and y represent the number of guitar lessons he attends.

a From this information, construct the set of five constraint inequalities.

b Graph the region defined by the five constraint inequalities.

c List the five vertices of the feasible region.

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d Using the five vertices determine the maximum number of guitar lessons that Edward can attend.

Objective functions

9 Write the objective function for each of the following situations:

a Students have set up a Mother's Day stall and are selling two products: photo frames for $\$5$ and necklaces for $\$12$. Write the objective function for the revenue earned, R, made by selling the items at the stall. Let x represent the number of photo frames sold, and let y represent the number of necklaces sold.

b A coffee cart sells two sizes of coffee: small for $\$3$ and large for $\$3.5$. Write the objective function for the revenue earned, R, made by the coffee cart selling the two products. Let x represent the number of small coffees sold, and let y represent the number of large coffees sold.

c A publisher hires two writers to write textbooks for them. Pauline writes 1100 words per hour and charges $\$25/\text{hr}$, and Tom writes 1150 words per hour and charges $\$30/\text{hr}$. Write the objective function for the cost, C, of hiring the two writers. Let x represent the number of hours that Pauline works for the publisher, and y represent the number of hours that Tom works for the publisher.

10 A factory produces two types of guitars: acoustics and electric. Each type of guitar requires a crafter and a tuner to make it. The time spent by each worker is given in the table.

	Crafter	Tuner
Acoustic Guitar	12 hours	11 hours
Electric Guitar	10 hours	9 hours
Max. work time per week	40 hours	35 hours

Let x represent the number of acoustic guitars built and y represent the number of electric guitars built.

a Construct the constraint inequalities for x and y given the number of guitars cannot be a negative number.

b Write an inequality relating x and y to the total time spent working by the crafter.

c Write an inequality relating x and y to the total time spent working by the tuner.

d If an acoustic guitar sells for $\$1400$ and an electric guitar sells for $\$900$, write the objective function for the revenue, R.

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